

DATA STRUCTURES

Course Code: CS-201

Credit Hours: 4(3+1)

Pre-requisite: CS-111

Course Objectives

The key objectives of this course include:

To understand the design of fundamental data structures as well as algorithms that operate on them.

To understand the fundamental tradeoffs in the design of the data structures.

To introduce tools for analyzing the time and space complexity of data structures.

Course Learning Outcomes

After successful completion of this course, the students should be able to: GA BT Level

1. Explain the fundamentals of data structure as well as algorithms that operate on them	2	C2
2. Apply and design various data structures methods for specified applications.	4	C3
3. Analyze empirically the behavior of algorithms in terms of time and memory used	3	C4

Course Contents

Abstract data types, complexity analysis, Big Oh notation, Arrays and its operations, linked list & its various types, Stacks (linked lists and array implementations), divide and conquer algorithms, Sorting algorithms (selection, insertion, merge, quick, bubble, heap, shell, radix, bucket), queue, dequeuer, priority queues (linked and array implementations of queues), binary search for sorted arrays, hashing and indexing, open addressing and chaining, trees and tree traversals, binary search trees, heaps, graphs, breadth-first and depth-first traversal, shortest path, adjacency matrix and adjacency list implementations.

Mapping of CLOs to GA's

GA's/CLOs	CLO1	CLO2	CLO3
GA1 (Academic Education)			
GA2 (Knowledge for Solving Computing Problems)	✓		
GA3 (Problem Analysis)			✓
GA4 (Design/Development of Solutions)		✓	
GA5 (Modern Tool Usage)			
GA6 (Individual and Teamwork)			
GA7 (Communication)			

GA8 (Computing Professionalism and Society)			
GA9 (Ethics)			
GA10 (Life-long Learning)			

Resources

Text Book:

Data Structures and Algorithms in C++ by Michael T. Goodrich, Roberto Tamassia, and David Mount, 4th Edition, 2014

Reference Books

Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss, 4th Edition), 2013

Data Structures in C++ by Cengage Learning, 8th Edition, 2018

Weekly Course Plan

Week No.	Topics
WEEK 01	Introduction to the course
	- Basic programming concepts - Introduction to Data Structures
WEEK 02	Abstract Data Types and Applications
	Analysis Tools - Complexity analysis - Asymptotic notations - Big Oh, Omega and theta notations
WEEK 03	Linear Arrays - Introduction to Linear Arrays - Memory addressing scheme in Linear Arrays - Algorithm for Insertion, searching, traversing and deletion in linear arrays
	Multidimensional Arrays - Multidimensional Array (2D array) - Memory addressing scheme in 2-D Array - Algorithm for addition and multiplication in 2-D array
WEEK 04	List
	- Reviving Structs and pointers - Introduction to Singly linked list - Memory addressing scheme of Singly linked list

	Algorithm for insertion, deletion, searching and traversing in singly linked list using pointers
WEEK 05	Algorithm for insertion, deletion, searching and traversing in singly linked list using pointers
	List - Introduction to Doubly linked list - Memory addressing scheme of Doubly linked list - Algorithm for insertion, deletion, searching and traversing in doubly linked list using pointers
WEEK 06	Circular linked list
	Stacks - Introduction to Stacks - Implementation of stack using arrays and linked list - Applications of stacks
WEEK 07	Infix to prefix and postfix conversion
	- Reversal of strings - Checking validity of brackets - Infix to postfix using stacks - Evaluation of postfix expression using stack
WEEK 08	Queues - Introduction to Queue data structure - Implementation of Queue using array and linked list - Insertion and deletion in linear queue
	- Insertion and deletion in circular queue - Priority queue and its types - Double ended queues
MID SEMESTER EXAM	
WEEK 10	Trees - Introduction to tree data structure - Basic terminologies related to tree - Representation of tree using array and linked list
	- Types of tree - Tree traversal techniques (Pre-order, In-order, Post-order)
WEEK 11	Trees

	• Formulation of binary tree from its traversals • Formulation of tree using infix expression
	• Introduction to Binary search tree (BST) • Insertion, deletion and searching in BST
WEEK 12	Heaps • Foundations, Binary Heaps • Binary Heaps, Heap Sort
	• Insertion Sort • Selection Sort
WEEK 13	Sorting • Merge Sort
	• Quick Sort
WEEK 14	Hashing • Introduction to hashing • Hash Functions
	• Chaining, Open Addressing
WEEK 15	Graphs • Basics Data Structures for Graphs
	Graph Traversals • Depth First Search • Breadth First Search
WEEK 16	Shortest Paths Algorithm • Dijkstra's Algorithm
	Minimum Spanning Trees • Kruskal's Algorithm • The Prim's Algorithm
END SEMESTER EXAMS	